

CS23532-COMPUTER NETWORKS-LAB MANUAL

Practical -1

AIM: - Study of various Network commands used in Linux and Windows:

Windows commands

```
C:\Users\Balag>hostname
```

BalaGanesh

```
C:\Users\Balag>ipconfig /all
```

Windows IP Configuration

Host Name : BalaGanesh

Primary Dns Suffix :

Node Type : Mixed

IP Routing Enabled. : No

WINS Proxy Enabled. : No

Ethernet adapter Ethernet 2:

Connection-specific DNS Suffix . :

Description : VirtualBox Host-Only Ethernet Adapter

Physical Address. : 0A-00-27-00-00-0E

DHCP Enabled. : No

Autoconfiguration Enabled : Yes

Link-local IPv6 Address : fe80::aa4a:775:1de1:1757%14(Preferred)

IPv4 Address. : 192.168.56.1(Preferred)

Subnet Mask : 255.255.255.0

Default Gateway :

DHCPv6 IAID : 118095911

DHCPv6 Client DUID. : 00-01-00-01-2D-79-FD-43-2C-58-B9-DE-5F-5F

NetBIOS over Tcpip. : Enabled

Wireless LAN adapter Local Area Connection* 1:

Media State : Media disconnected

Connection-specific DNS Suffix . :

Description : Microsoft Wi-Fi Direct Virtual Adapter

Physical Address. : F8-FE-5E-3A-70-56

DHCP Enabled. : Yes

Autoconfiguration Enabled : Yes

Wireless LAN adapter Local Area Connection* 2:

Media State : Media disconnected

Connection-specific DNS Suffix . :

Description : Microsoft Wi-Fi Direct Virtual Adapter #2

Physical Address. : FA-FE-5E-3A-70-55

DHCP Enabled. : No

Autoconfiguration Enabled : Yes

Wireless LAN adapter Wi-Fi:

Connection-specific DNS Suffix . :

Description : Intel(R) Wi-Fi 6E AX211 160MHz

Physical Address. : F8-FE-5E-3A-70-55

DHCP Enabled. : Yes
Autoconfiguration Enabled : Yes
Link-local IPv6 Address : fe80::b810:29c9:55e9:3a65%17(Preferred)
IPv4 Address. : 10.83.96.83(Preferred)
Subnet Mask : 255.255.255.0
Lease Obtained. : 26 August 2026 14:38:14
Lease Expires : 26 August 2026 15:38:11
Default Gateway : 10.83.96.20
DHCP Server : 10.83.96.20
DHCPv6 IAID : 167312990
DHCPv6 Client DUID. : 00-01-00-01-2D-79-FD-43-2C-58-B9-DE-5F-5F
DNS Servers : 10.83.96.20
NetBIOS over Tcpi. : Enabled

Ethernet adapter Bluetooth Network Connection:

Media State : Media disconnected
Connection-specific DNS Suffix . :
Description : Bluetooth Device (Personal Area Network)
Physical Address. : F8-FE-5E-3A-70-59
DHCP Enabled. : Yes
Autoconfiguration Enabled : Yes

Ethernet adapter Ethernet:

Media State : Media disconnected
Connection-specific DNS Suffix . :
Description : Realtek Gaming GbE Family Controller

Physical Address. : 2C-58-B9-DE-5F-5F

DHCP Enabled. : Yes

Autoconfiguration Enabled : Yes

C:\Users\Balag>arp -a

Interface: 192.168.56.1 --- 0xe

Internet Address	Physical Address	Type
192.168.56.255	ff-ff-ff-ff-ff	static
224.0.0.22	01-00-5e-00-00-16	static
224.0.0.251	01-00-5e-00-00-fb	static
224.0.0.252	01-00-5e-00-00-fc	static
239.255.255.250	01-00-5e-7f-ff-fa	static
255.255.255.255	ff-ff-ff-ff-ff	static

Interface: 10.83.96.83 --- 0x11

Internet Address	Physical Address	Type
10.83.96.20	4e-ec-85-ec-ff-15	dynamic
10.83.96.255	ff-ff-ff-ff-ff	static
224.0.0.22	01-00-5e-00-00-16	static
224.0.0.251	01-00-5e-00-00-fb	static
224.0.0.252	01-00-5e-00-00-fc	static
239.255.255.250	01-00-5e-7f-ff-fa	static
255.255.255.255	ff-ff-ff-ff-ff	static

C:\Users\Balag>netstat -r

=====

Interface List

14...0a 00 27 00 00 0eVirtualBox Host-Only Ethernet Adapter
11...f8 fe 5e 3a 70 56Microsoft Wi-Fi Direct Virtual Adapter
4...fa fe 5e 3a 70 55Microsoft Wi-Fi Direct Virtual Adapter #2
17...f8 fe 5e 3a 70 55Intel(R) Wi-Fi 6E AX211 160MHz
19...f8 fe 5e 3a 70 59Bluetooth Device (Personal Area Network)
2...2c 58 b9 de 5f 5fRealtek Gaming GbE Family Controller
1.....Software Loopback Interface 1

IPv4 Route Table

Active Routes:

Network	Destination	Netmask	Gateway	Interface	Metric
	0.0.0.0	0.0.0.0	10.83.96.20	10.83.96.83	35
	10.83.96.0	255.255.255.0	On-link	10.83.96.83	291
	10.83.96.83	255.255.255.255	On-link	10.83.96.83	291
	10.83.96.255	255.255.255.255	On-link	10.83.96.83	291
	127.0.0.0	255.0.0.0	On-link	127.0.0.1	331
	127.0.0.1	255.255.255.255	On-link	127.0.0.1	331
	127.255.255.255	255.255.255.255	On-link	127.0.0.1	331
	192.168.56.0	255.255.255.0	On-link	192.168.56.1	281
	192.168.56.1	255.255.255.255	On-link	192.168.56.1	281
	192.168.56.255	255.255.255.255	On-link	192.168.56.1	281
	224.0.0.0	240.0.0.0	On-link	127.0.0.1	331
	224.0.0.0	240.0.0.0	On-link	192.168.56.1	281
	224.0.0.0	240.0.0.0	On-link	10.83.96.83	291
	255.255.255.255	255.255.255.255	On-link	127.0.0.1	331
	255.255.255.255	255.255.255.255	On-link	192.168.56.1	281

255.255.255.255 255.255.255.255 On-link 10.83.96.83 291

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Persistent Routes:

None

IPv6 Route Table

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Active Routes:

If	Metric	Network	Destination	Gateway
----	--------	---------	-------------	---------

1	331	::1/128		On-link
---	-----	---------	--	---------

14	281	fe80::/64		On-link
----	-----	-----------	--	---------

17	291	fe80::/64		On-link
----	-----	-----------	--	---------

14	281	fe80::aa4a:775:1de1:1757/128		
----	-----	------------------------------	--	--

On-link

17	291	fe80::b810:29c9:55e9:3a65/128		
----	-----	-------------------------------	--	--

On-link

1	331	ff00::/8		On-link
---	-----	----------	--	---------

14	281	ff00::/8		On-link
----	-----	----------	--	---------

17	291	ff00::/8		On-link
----	-----	----------	--	---------

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Persistent Routes:

None

C:\Users\Balag>nslookup www.google.com

Server: UnKnown

Address: 10.83.96.20

Non-authoritative answer:

Name: www.google.com

Addresses: 2001:4860:482d:7700::

2001:4860:4826:7700::

2001:4860:4828:7700::

2001:4860:482b:7700::

2001:4860:4827:7700::

2001:4860:4829:7700::

2001:4860:482a:7700::

2001:4860:482c:7700::

142.251.152.119

142.251.156.119

142.251.154.119

142.251.151.119

142.251.153.119

142.251.157.119

142.251.150.119

142.251.155.119

C:\Users\Balag>ping google.com

Pinging google.com [142.251.43.46] with 32 bytes of data:

Reply from 142.251.43.46: bytes=32 time=14ms TTL=114

Reply from 142.251.43.46: bytes=32 time=233ms TTL=114

Reply from 142.251.43.46: bytes=32 time=59ms TTL=114

Reply from 142.251.43.46: bytes=32 time=328ms TTL=114

Ping statistics for 142.251.43.46:

Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),

Approximate round trip times in milli-seconds:

Minimum = 14ms, Maximum = 328ms, Average = 158ms

C:\Users\Balag>route print

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Interface List

14...0a 00 27 00 00 0eVirtualBox Host-Only Ethernet Adapter
11...f8 fe 5e 3a 70 56Microsoft Wi-Fi Direct Virtual Adapter
4...fa fe 5e 3a 70 55Microsoft Wi-Fi Direct Virtual Adapter #2
17...f8 fe 5e 3a 70 55Intel(R) Wi-Fi 6E AX211 160MHz
19...f8 fe 5e 3a 70 59Bluetooth Device (Personal Area Network)
2...2c 58 b9 de 5f 5fRealtek Gaming GbE Family Controller
1.....Software Loopback Interface 1

=====

IPv4 Route Table

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Active Routes:

Network	Destination	Netmask	Gateway	Interface	Metric
0.0.0.0	0.0.0.0	10.83.96.20	10.83.96.83	35	
10.83.96.0	255.255.255.0	On-link	10.83.96.83	291	
10.83.96.83	255.255.255.255	On-link	10.83.96.83	291	
10.83.96.255	255.255.255.255	On-link	10.83.96.83	291	
127.0.0.0	255.0.0.0	On-link	127.0.0.1	331	
127.0.0.1	255.255.255.255	On-link	127.0.0.1	331	
127.255.255.255	255.255.255.255	On-link	127.0.0.1	331	
192.168.56.0	255.255.255.0	On-link	192.168.56.1	281	

192.168.56.1	255.255.255.255	On-link	192.168.56.1	281
192.168.56.255	255.255.255.255	On-link	192.168.56.1	281
224.0.0.0	240.0.0.0	On-link	127.0.0.1	331
224.0.0.0	240.0.0.0	On-link	192.168.56.1	281
224.0.0.0	240.0.0.0	On-link	10.83.96.83	291
255.255.255.255	255.255.255.255	On-link	127.0.0.1	331
255.255.255.255	255.255.255.255	On-link	192.168.56.1	281
255.255.255.255	255.255.255.255	On-link	10.83.96.83	291

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Persistent Routes:

None

IPv6 Route Table

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Active Routes:

If	Metric	Network	Destination	Gateway
----	--------	---------	-------------	---------

1	331	::1/128		On-link
14	281	fe80::/64		On-link
17	291	fe80::/64		On-link
14	281	fe80::aa4a:775:1de1:1757/128		

On-link

17	291	fe80::b810:29c9:55e9:3a65/128		
----	-----	-------------------------------	--	--

On-link

1	331	ff00::/8		On-link
14	281	ff00::/8		On-link
17	291	ff00::/8		On-link

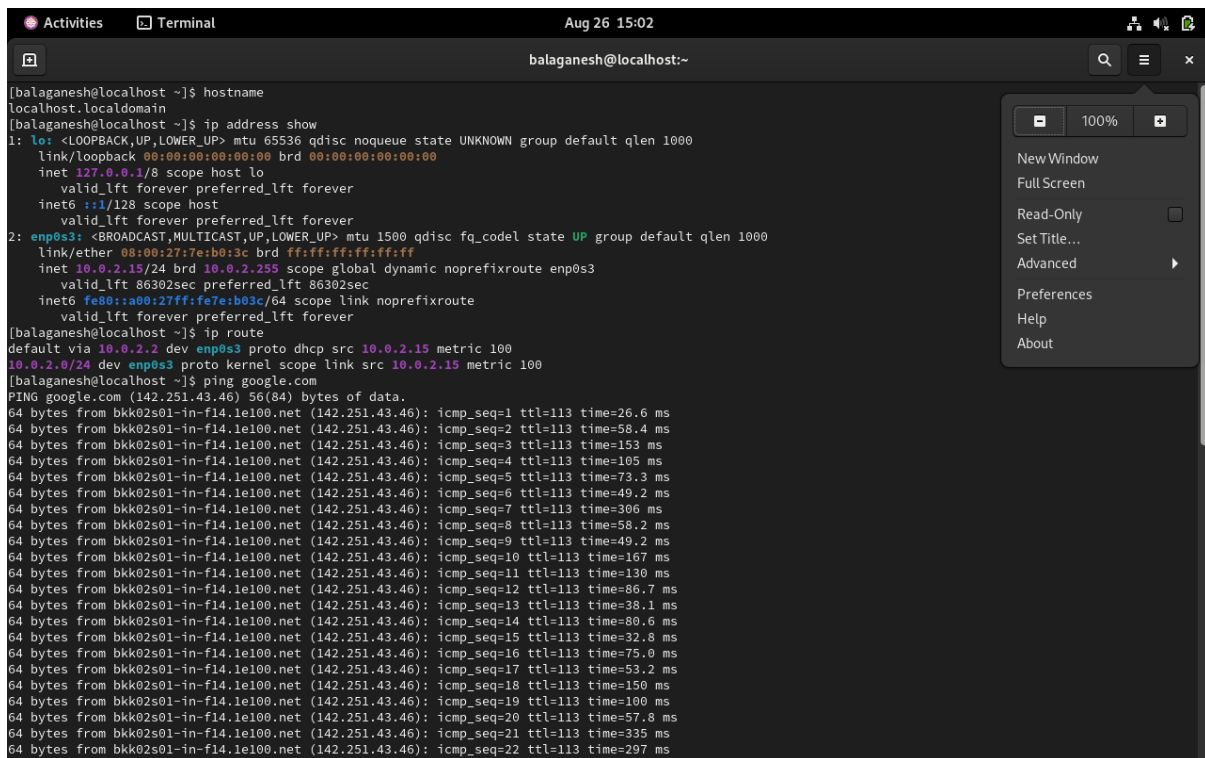
=====

Persistent Routes:

None

C:\Users\Balag>tracert google.com

Linux commands



The screenshot shows a terminal window titled "balaganesh@localhost:~" with the following commands and output:

```
[balaganesh@localhost ~]$ hostname
localhost.localdomain
[balaganesh@localhost ~]$ ip address show
1: lo: <LOOPBACK,UP,LOWER_UP> mtu 65536 qdisc noqueue state UNKNOWN group default qlen 1000
    link/loopback 00:00:00:00:00:00 brd 00:00:00:00:00:00
    inet 127.0.0.1/8 scope host lo
        valid_lft forever preferred_lft forever
    inet6 ::1/128 scope host
        valid_lft forever preferred_lft forever
2: enp0s3: <BROADCAST,MULTICAST,UP,LOWER_UP> mtu 1500 qdisc fq_codel state UP group default qlen 1000
    link/ether 08:00:27:7e:b0:3c brd ff:ff:ff:ff:ff:ff
    inet 10.0.2.15/24 brd 10.0.2.255 scope global dynamic noprefixroute enp0s3
        valid_lft 86302sec preferred_lft 86302sec
    inet6 fe80::a00:27ff:fe7e:b03c/64 scope link noprefixroute
        valid_lft forever preferred_lft forever
[balaganesh@localhost ~]$ ip route
default via 10.0.2.2 dev enp0s3 proto dhcp src 10.0.2.15 metric 100
10.0.2.0/24 dev enp0s3 proto kernel scope link src 10.0.2.15 metric 100
[balaganesh@localhost ~]$ ping google.com
PING google.com (142.251.43.46) 56(84) bytes of data.
64 bytes from bkk02s01-in-f14.1e100.net (142.251.43.46): icmp_seq=1 ttl=113 time=26.6 ms
64 bytes from bkk02s01-in-f14.1e100.net (142.251.43.46): icmp_seq=2 ttl=113 time=58.4 ms
64 bytes from bkk02s01-in-f14.1e100.net (142.251.43.46): icmp_seq=3 ttl=113 time=153 ms
64 bytes from bkk02s01-in-f14.1e100.net (142.251.43.46): icmp_seq=4 ttl=113 time=105 ms
64 bytes from bkk02s01-in-f14.1e100.net (142.251.43.46): icmp_seq=5 ttl=113 time=73.3 ms
64 bytes from bkk02s01-in-f14.1e100.net (142.251.43.46): icmp_seq=6 ttl=113 time=49.2 ms
64 bytes from bkk02s01-in-f14.1e100.net (142.251.43.46): icmp_seq=7 ttl=113 time=306 ms
64 bytes from bkk02s01-in-f14.1e100.net (142.251.43.46): icmp_seq=8 ttl=113 time=58.2 ms
64 bytes from bkk02s01-in-f14.1e100.net (142.251.43.46): icmp_seq=9 ttl=113 time=49.2 ms
64 bytes from bkk02s01-in-f14.1e100.net (142.251.43.46): icmp_seq=10 ttl=113 time=167 ms
64 bytes from bkk02s01-in-f14.1e100.net (142.251.43.46): icmp_seq=11 ttl=113 time=130 ms
64 bytes from bkk02s01-in-f14.1e100.net (142.251.43.46): icmp_seq=12 ttl=113 time=86.7 ms
64 bytes from bkk02s01-in-f14.1e100.net (142.251.43.46): icmp_seq=13 ttl=113 time=38.1 ms
64 bytes from bkk02s01-in-f14.1e100.net (142.251.43.46): icmp_seq=14 ttl=113 time=80.6 ms
64 bytes from bkk02s01-in-f14.1e100.net (142.251.43.46): icmp_seq=15 ttl=113 time=32.8 ms
64 bytes from bkk02s01-in-f14.1e100.net (142.251.43.46): icmp_seq=16 ttl=113 time=75.0 ms
64 bytes from bkk02s01-in-f14.1e100.net (142.251.43.46): icmp_seq=17 ttl=113 time=53.2 ms
64 bytes from bkk02s01-in-f14.1e100.net (142.251.43.46): icmp_seq=18 ttl=113 time=150 ms
64 bytes from bkk02s01-in-f14.1e100.net (142.251.43.46): icmp_seq=19 ttl=113 time=100 ms
64 bytes from bkk02s01-in-f14.1e100.net (142.251.43.46): icmp_seq=20 ttl=113 time=57.8 ms
64 bytes from bkk02s01-in-f14.1e100.net (142.251.43.46): icmp_seq=21 ttl=113 time=335 ms
64 bytes from bkk02s01-in-f14.1e100.net (142.251.43.46): icmp_seq=22 ttl=113 time=297 ms
```

Student observation answers

1. **Command used to check reachability:** ping
2. **Command used to find hops:** tracert in Windows / traceroute in Linux
3. **Command to display IP configuration:** ipconfig in Windows / ip address in Linux
4. **Command to display TCP port status:** netstat
5. **Command to modify IP configuration in Linux:** ip address / ip route